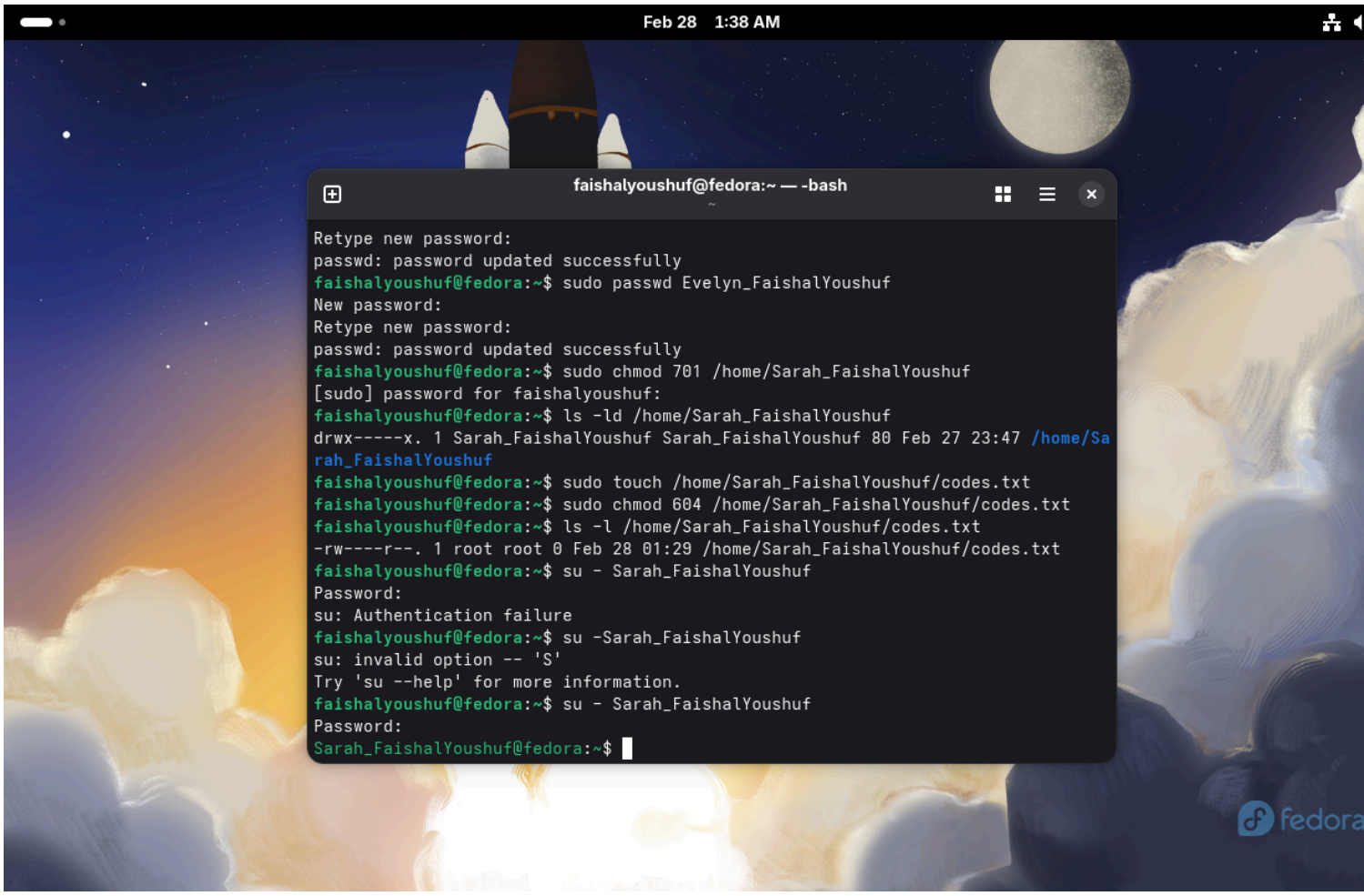
1.

a) given the snapshot:



b)So, if Evelyn reboots the system, she could access the default system files and easily get into Sarah_FaishalYoushuf's home directory as a root user. In this case, Linux directory structures typically continue to allow getting access if the files are heavily mounted. So, without complete disk authentication, Evelyn can read the text file and get unauthorized permission.

Feb 28 1:44 AM

faishalyoushuf@fedora:~ — -bash

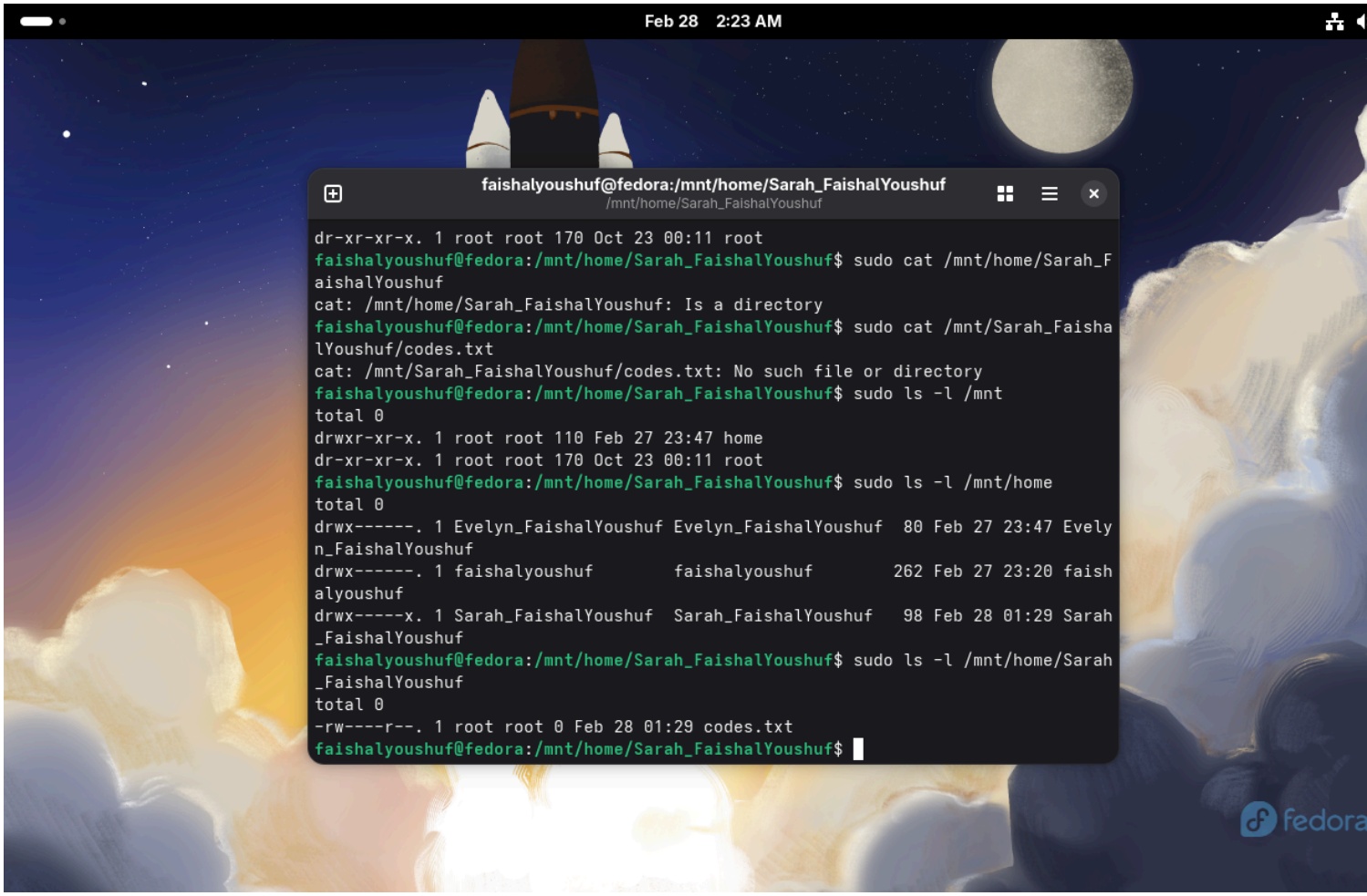
```
faishalyoushuf@fedora:~$ su -Sarah_FaishalYoushuf
su: invalid option -- 'S'
Try 'su --help' for more information.
faishalyoushuf@fedora:~$ su - Sarah_FaishalYoushuf
Password:
Sarah_FaishalYoushuf@fedora:~$ sudo -u Evelyn_FaishalYoushuf cat /home/Sarah_Fai
shalYoushuf/codes.txt

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for Sarah_FaishalYoushuf:
Sarah_FaishalYoushuf is not in the sudoers file.
Sarah_FaishalYoushuf@fedora:~$ sudo -u Evelyn_FaishalYoushuf ls /home/Sarah_Fais
halYoushuf
[sudo] password for Sarah_FaishalYoushuf:
Sarah_FaishalYoushuf is not in the sudoers file.
Sarah_FaishalYoushuf@fedora:~$
```

Feb 28 2:11 AM

faishalyoushuf@fedora:~

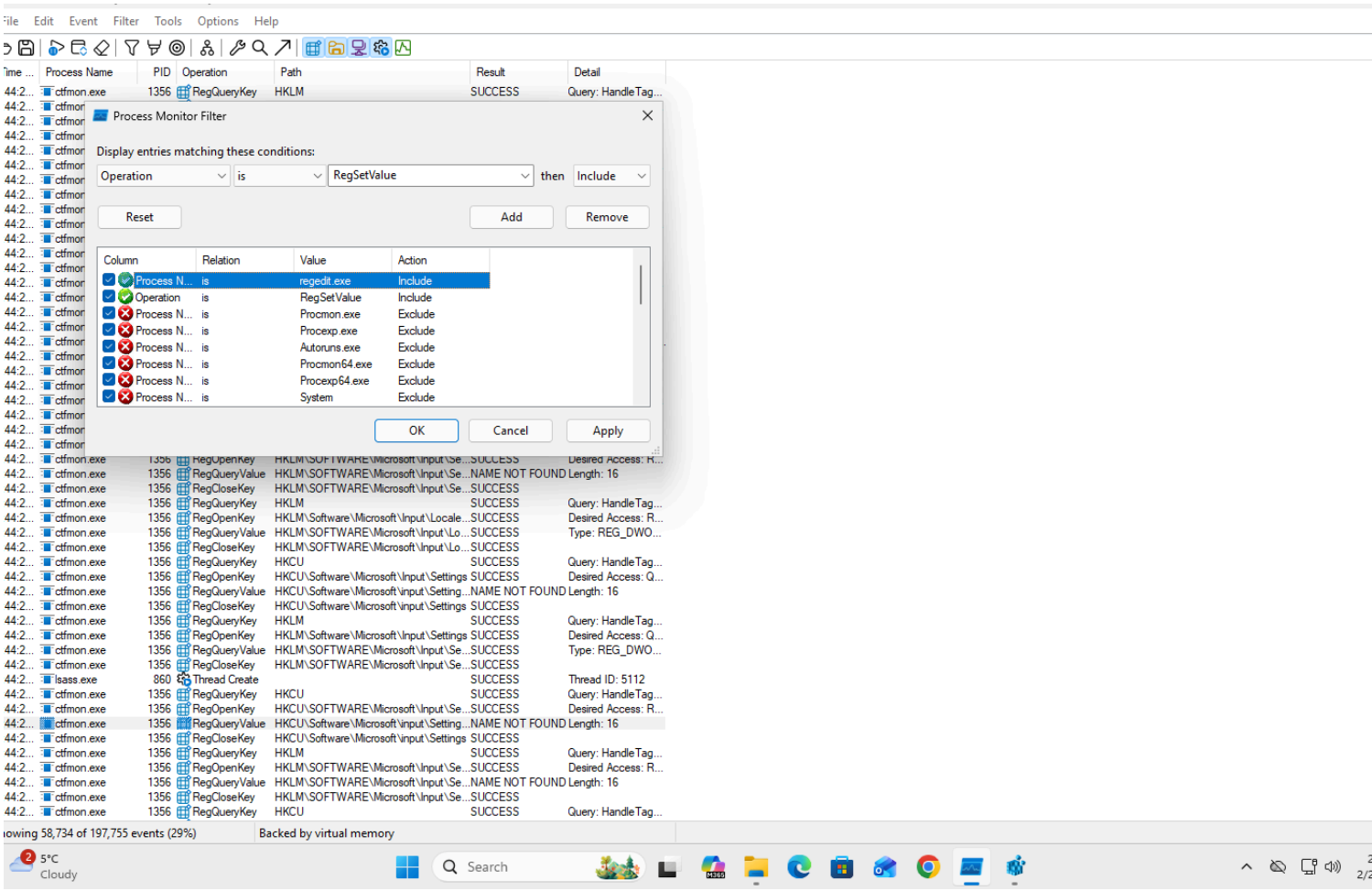
```
drwxr-xr-x. 1 faishalyoushuf faishalyoushuf 0 Feb 27 19:33 Music
drwxr-xr-x. 1 faishalyoushuf faishalyoushuf 0 Feb 27 19:33 Pictures
drwxr-xr-x. 1 faishalyoushuf faishalyoushuf 0 Feb 27 19:33 Public
drwxr-xr-x. 1 faishalyoushuf faishalyoushuf 0 Feb 27 19:33 Templates
drwxr-xr-x. 1 faishalyoushuf faishalyoushuf 0 Feb 27 19:33 Videos
faishalyoushuf@fedora:~$ cat codes.txt
cat: codes.txt: No such file or directory
faishalyoushuf@fedora:~$ ls -l cat codes.txt
ls: cannot access 'cat': No such file or directory
ls: cannot access 'codes.txt': No such file or directory
faishalyoushuf@fedora:~$ ^C
faishalyoushuf@fedora:~$ sudo whoami
root
faishalyoushuf@fedora:~$ lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   15G  0 disk
├─sda1   8:1    0  600M  0 part /mnt
│                               /boot/efi
├─sda2   8:2    0    2G  0 part /boot
└─sda3   8:3    0 12.4G  0 part /home
                                /
sr0     11:0    1  2.4G  0 rom  /run/media/faishalyoushuf/Fedora-WS-Live-43
zram0  251:0    0  3.2G  0 disk [SWAP]
faishalyoushuf@fedora:~$
```

c) If Sarah encodes the files but fails to assign proper file permissions, another individual might access the files that were encrypted and attempt to figure out her username and password by use of brute force. An attacker might additionally eliminate or modify the files that were encrypted, and this would cause an interruption of service conditions or manipulation. So, encryption alone won't be sufficient without proper file permissions.



2.

On a Windows system, when we run various filtering options in Process Monitor, we added Filter by operation, Filter by Process Name, and Filter by Registry Path and Exclusion Filter. Filter By Operation it restrits to only registry value write operations. Filter by process name here result is changed by windows registry editor. Filter By Registry Path HKCU\Software\Microsoft\Windows\CurrentVersion\Run is very sensitive to being attacked.

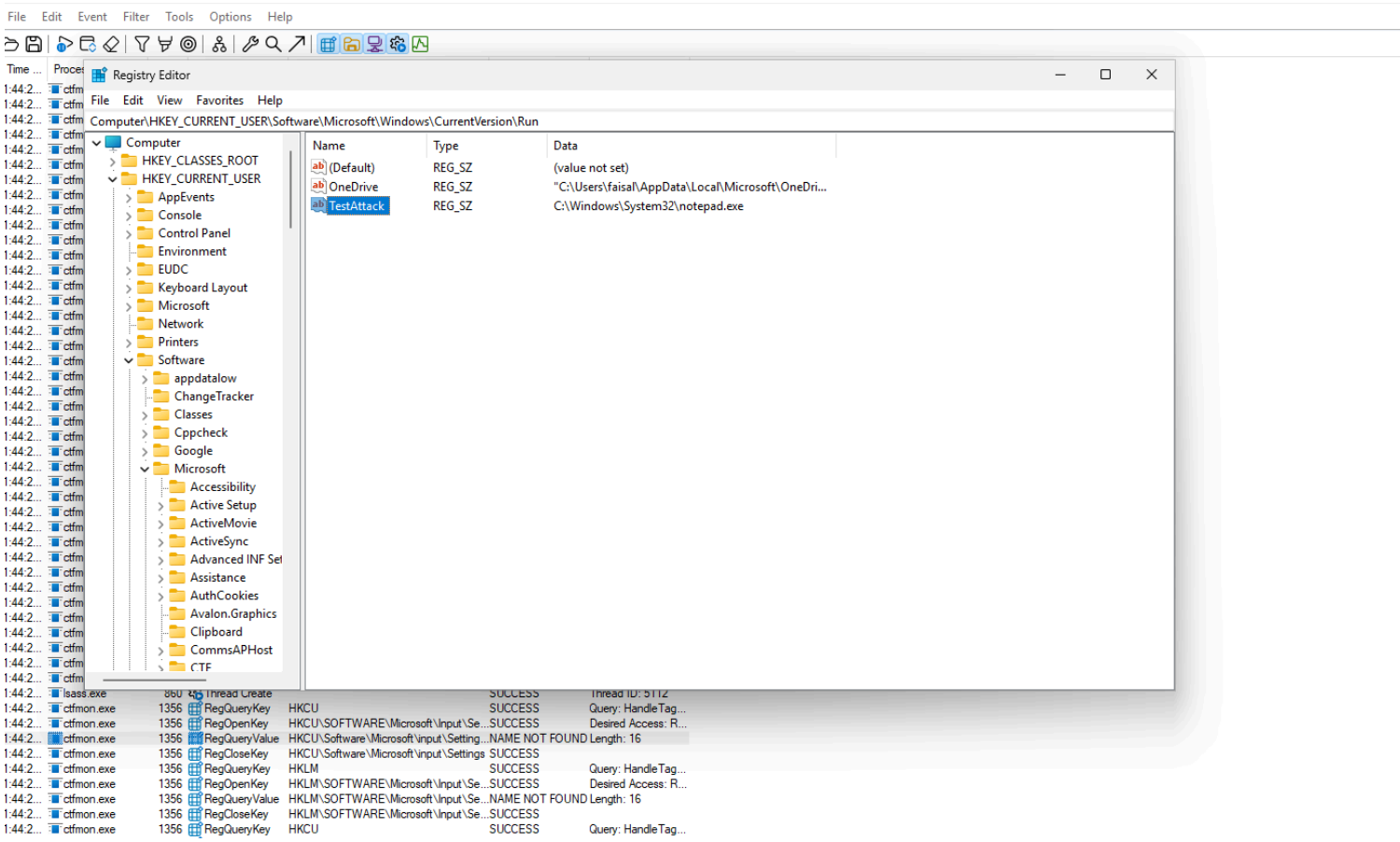


The exclusion filter also proves how processes can be reduced by using ProcMon.

2a) Software\Microsoft\Windows\Current Version\Run

by including a new value leading to a malicious executable like malware.exe. This is accomplished by the malicious software running on each occasion the operating system initiates, because the programs in the executable key run themselves when the user signs in. This gives the attacker drive, so they are able to log in after the restart.

2b) Here I have reset all the filters and then included 3 filters following the 1 exclude, and then following the path and after multiple occasions, in Windows Registry Editor, it showed success.



Following Hkey_current_user to run, at TestAttack, I got the path, and then in Process Monitor, in the filtering options, it showed success as attached before.

Filter by operation:

Column: process operation condition: is Value: RegSetValue Action: include

Filter By Operation it restrits to only registry value write operations.

Column: process path condition: contains Value: CurrentVersion\Run Action: include

Filter by process name here result is changed by windows registry editor.

Column: process name, condition: is Value: Regedit.exe Action: include

Filter By Registry Path HKCU\Software\Microsoft\Windows\CurrentVersion\Run is very sensitive to being attacked.

Column: process name, condition: is Value: FaishalYoushuf Action: exclude

The exclusion filter also proves how processes can be reduced by using ProcMon. It helps to filter from unauthrosied access for detection.